25901422 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCPEG200 - ATF AND LATEST DEVELOPMENT IN FUEL TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. Identify the primary objective of integrating biofuel production into biorefineries. [U] / [CO1]

   a). Maximum utilization of biomass b). Single-product formation

   c). Increased fossil fuel use d). Elimination of co-products

2. Write the suitable technology when multiple co-products (corn oil, gluten meal) are required. [A] / [CO2]

   a). Dry milling b). Wet milling

   c). Simple distillation d). Anaerobic digestion

3. Compare ethanol and gasoline based on calorific value. [U] / [CO2]

   a). Ethanol has higher calorific value than gasoline

   b). Ethanol has lower calorific value than gasoline

   c). Both have the same calorific value

d). Calorific value is not related to fuels

4. The primary purpose of filtration of aviation fuel before refuelling is to: [K] / [CO3]

a). Improve calorific value

b). Remove suspended solids and rust particles

c). Increase volatility

d). Reduce freezing point

5. Which of the following aviation fuels has a high flash point and low volatility? [U] / [CO4]

a). AV Gas b). Broad-fraction fuel

c). Aviation kerosene (Jet fuel) d). Automotive gasoline

6. In the ATJ process, aviation fuel is produced from: [K] / [CO4]

a). Crude oil b). Syngas

c). Alcohols derived from biomass d). Animal fats

7. Motor Octane Number (MON) is measured under: [K] / [CO4]

a). Mild engine conditions b). Severe operating conditions

c). Low temperature conditions d). Idle conditions

8. Low electrical conductivity of ATF may lead to: [K] / [CO5]

a). Poor combustion b). Static electricity accumulation

c). High freezing point d). Corrosion

9. Excess aromatic content in ATF leads to: [U] / [CO5]

a). Increased smoke formation b). Reduced smoke

c). Increased thermal stability d). Decreased smoke formation

10. ASTM D7566 is related to: [K] / [CO6]

a). Fuel storage stability b). Certification of aviation biofuels

c). Engine design d). Emission control

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11.	Explain the importance of waste utilization in biofuel production.	[U] / [CO1]
12.	Define lubricity and explain why lubricity additives are needed in modern jet fuels.	[K] / [CO4]
13.	Define aviation fuels and state their main classification based on engine type.	[K] / [CO3]
14.	Discuss on total additive sulphur in aviation fuels.	[U] / [CO4]
15.	State any three chemical properties of Aviation Turbine Fuel.	[K] / [CO5]
16.	Examine the limitations of hydrogen as an aviation fuel.	[An] / [CO6]

[3x4 = 12]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

17.	Outline anaerobic digestion for biogas production.	[K] / [CO1]
18.	Illustrate a suitable production technology for an ethanol plant designed for low capital investment and simpler operation.	[A] / [CO2]
19.	Identify the advantages of AV gas fuel over turbine engine fuel.	[K] / [CO3]
20.	List the advantages and limitations of using SAF in modern aviation.	[K] / [CO3]
21.	State the hydrotreating process used in the manufacture of ATF, including operating conditions, reactions involved, and improvements in fuel quality	[K] / [CO5]
22.	Describe the hydrodesulphurization process used in the production of Aviation Turbine Fuel, highlighting its role in improving fuel quality.	[U] / [CO5]

[7x4 = 28]